

Cryptography and Network Security

Integrated Situation — Technical Report

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1 Introduction

This report documents the security assessment and technical work carried out for the ETTC801 Integrated Situation. The scenario involves a polytechnic institute storing student records on a central server and transferring files between two campuses, with weak staff passwords, outdated software, unencrypted transfers, and guest network access to the records server identified as issues in a security review.

2 Risk Assessment

2.1 Assets, Vulnerabilities and Consequences

Asset	Vulnerability	Consequence
Central student records server	Guest network can reach the server	Unauthorised access, theft or tampering
Staff login credentials	Weak/reused passwords	Account compromise, unauthorised access
Inter-campus file transfer channel	Transfers are unencrypted	Interception (eavesdropping) in transit

Table 1: Assets, vulnerabilities and consequences

2.2 Risk Ranking

Rank	Risk	Likelihood	Reasoning
1	Guest access to records server	High	No segmentation exists; repeated probing already observed from an unfamiliar external address.
2	Weak staff passwords	Medium–High	Easy to guess/brute-force; grants legitimate-looking access.
3	Unencrypted transfers	Medium	Requires attacker on the network path, but still a real confidentiality risk.

Table 2: Risk ranking by likelihood and impact

2.3 Recommended Controls

- Guest access to records server: network segmentation (VLANs) and firewall rules denying guest traffic
- Weak staff passwords: strong password policy plus multi-factor authentication (MFA)
- Unencrypted transfers: encrypt in transit (SFTP/SCP or TLS) instead of plaintext protocols

3 Encryption and Integrity Toolkit

The `crypto_toolkit.py` programme was developed to encrypt and decrypt student records and to verify their integrity.

3.1 Design

- **Encryption:** symmetric encryption using Fernet (AES-128 in CBC mode with HMAC authentication) from the Python `cryptography` library.
- **Integrity:** SHA-256 hashing via `hashlib`, with the hash saved alongside the file and re-checked on demand.
- **Robustness:** all file operations are wrapped in `try/except` blocks so missing files or invalid input do not crash the programme.

3.2 Test Evidence

```
$ python3 crypto_toolkit.py genkey
[OK] Key generated and saved to secret.key

$ python3 crypto_toolkit.py encrypt sample_record.txt sample_record.enc
[OK] Encrypted 'sample_record.txt' -> 'sample_record.enc'

$ python3 crypto_toolkit.py hash sample_record.enc
[OK] SHA-256 of 'sample_record.enc':
    69fe15c94f975f06fc1ac371ae85c43c37346bfdbf982416ad2bc9e28992962c

$ python3 crypto_toolkit.py verify sample_record.enc sample_record.enc.
sha256
[OK] Integrity check PASSED - file has not been modified.

$ python3 crypto_toolkit.py decrypt sample_record.enc sample_record.dec
[OK] Decrypted 'sample_record.enc' -> 'sample_record.dec'
$ diff sample_record.txt sample_record.dec
MATCH: decrypted file is identical to original

$ echo "tampered" >> sample_record.enc
$ python3 crypto_toolkit.py verify sample_record.enc sample_record.enc.
sha256
[ALERT] Integrity check FAILED - file has been modified!
```

The final test confirms the toolkit correctly detects tampering after the encrypted file was deliberately modified.

4 Network Traffic Filtering

Firewall rules were applied in the authorised laboratory environment using `iptables` (see `firewall.rules.sh`), tested with a local HTTP service on port 8080 and three simulated source subnets (staff, guest, external).

4.1 Rules Summary

- Block guest subnet (192.168.20.0/24) → service port
- Allow staff subnet (192.168.10.0/24) → service port
- Drop all other inbound traffic to that port

4.2 Test Procedure and Results

Test	Source	Expected	Result
1	Staff subnet (192.168.10.5)	Allowed	Succeeded
2	Guest subnet (192.168.20.5)	Blocked	Timed out
3	External (172.16.0.5)	Blocked	Timed out

Table 3: Firewall test results

```
$ nc -vz -s 192.168.10.5 127.0.0.1 8080
Connection to 127.0.0.1 8080 port [tcp/http-alt] succeeded!

$ nc -vz -s 192.168.20.5 127.0.0.1 8080
nc: connect to 127.0.0.1 port 8080 (tcp) failed: Connection timed out

$ nc -vz -s 172.16.0.5 127.0.0.1 8080
nc: connect to 127.0.0.1 port 8080 (tcp) failed: Connection timed out
```

5 Conclusion

The measures implemented – network segmentation via firewall rules, authenticated symmetric encryption, and SHA-256 integrity verification – address the vulnerabilities identified in the risk assessment. Testing confirmed that staff access is permitted while guest and external access to the protected service are correctly blocked, and that the encryption toolkit correctly encrypts, decrypts, and detects tampering.

GitHub repository: <https://github.com/josue-03/cryptography-network-security-exam>